

Supplier Delay Prediction

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1. Executive Summary

A manufacturing company is experiencing supplier delivery issues that can disrupt production schedules, increase inventory costs, and reduce customer service performance. This report uses supervised predictive analytics by building two models: a linear regression model to predict how many days late or early a purchase order will arrive, and a logistic regression model to predict whether a purchase order will be delayed or on time.

The dataset includes 1,200 purchase orders. Overall, 953 purchase orders were delayed, or about 79.42% of the dataset. Only 247 purchase orders were on time, or about 20.58%. This shows that supplier delays are a major operational issue for the company.

The analysis makes sure to avoid data leakage. For the linear regression model predicting Delay_Days, the variables Delay_Status and Actual_Delivery_Days were excluded as they are known after the outcome. For the logistic regression model predicting Delay_Status, the variables Delay_Days and Actual_Delivery_Days were excluded. The ID variables PO_ID and Supplier_ID were also removed as they are used as row identifiers rather than explaining supplier performance.

The linear regression model had a test MAE of about 2.80 days and a test RMSE of about 3.54 days. This means the model's predictions were typically off by about three days. The logistic regression model had an accuracy of about 85.28%, which improved over the baseline accuracy of about 79.44%. The model was strong at identifying delayed orders, but it was weaker at correctly identifying on time orders.

2. Business Problem and Objective

The company's main business problem is that supplier purchase orders are often not arriving as expected. In manufacturing, supplier delays can create serious problems. Due to this, the company wants to use predictive analytics to help planning decisions before the delivery occurs. If managers can predict the number of delay days or the probability of a delay, they can take action earlier.

The first objective is to predict Delay_Days. This is a regression problem as the target variable is numeric. A positive value means the order arrived late, while a negative value means the order arrived early. For example, a prediction of 5 would mean the model expects the order to arrive about five days late, while a prediction of -2 would mean the order is expected to arrive about two days early.

The second objective is to predict Delay_Status. This is a classification problem because the target variable is categorical. Each order is classified as either Delayed or On_Time. This model helps answer whether a purchase order is likely to be delayed or on time.

Both models are useful for management. The linear regression model gives a more detailed estimate of timing risk, while the logistic regression model gives a simpler delayed versus on time classification. Together, these models can help the company adjust production schedules and manage risk before a purchase order is completed.

3. Dataset and Variables Description

The original dataset contains 1,200 purchase orders and 24 variables. The working dataset kept only the variables that are appropriate for prediction before

delivery occurs. The variables used as predictors included Supplier_Region, Supplier_Rating, Product_Category, Material_Criticality, Season, Order_Quantity, Unit_Cost, Order_Value, Lead_Time_Days, Distance_Miles, Expedited, Rush_Order, Supplier_Years, Past_Orders_Count, Prior_Delay_Rate, On_Time_Rate, Defect_Rate, Quality_Audit_Score, and Order_Complexity.

The two target variables were Delay_Days and Delay_Status. The average value of Delay_Days was about 5.05 days, meaning that the average order arrived about five days late. The minimum value was -5 days, showing that some orders arrived early, while the maximum value was 24 days, showing that some orders were significantly late.

The target distribution for Delay_Status was highly unbalanced. There were 953 delayed orders and 247 on-time orders. This means about 79.42% of orders were delayed, while only 20.58% were on time.

The descriptive analysis showed clear differences between delayed and on-time orders. Delayed orders had an average delay of about 6.88 days, while on-time orders arrived about 2.00 days early on average. Delayed orders also had lower average supplier ratings, higher prior delay rates, lower on-time rates, higher defect rates, lower quality audit scores, and higher order complexity.

4. Linear Regression Results

The linear regression model was built to predict Delay_Days, which measures how many days late or early a purchase order arrived. Since this is a numeric target

variable, linear regression is appropriate. The model used a 70/30 train-test split, with 840 observations used for training and 360 observations used for testing.

The linear regression model had an R-squared of about 0.574 on the training data and an adjusted R-squared of about 0.559. This means the model explained a little more than half of the variation in delay days using the selected supplier and order predictors. For a business dataset with many operational factors, this is a reasonably useful result.

On the test data, the model had a mean absolute error of about 2.80 days and a root mean squared error of about 3.54 days. The MAE means that, on average, the model's prediction was off by about 2.8 days. The RMSE is higher because it penalizes larger prediction errors more strongly. This suggests that the model can provide a useful estimate of delivery timing, but managers should treat the prediction as a planning estimate rather than an exact delivery date.

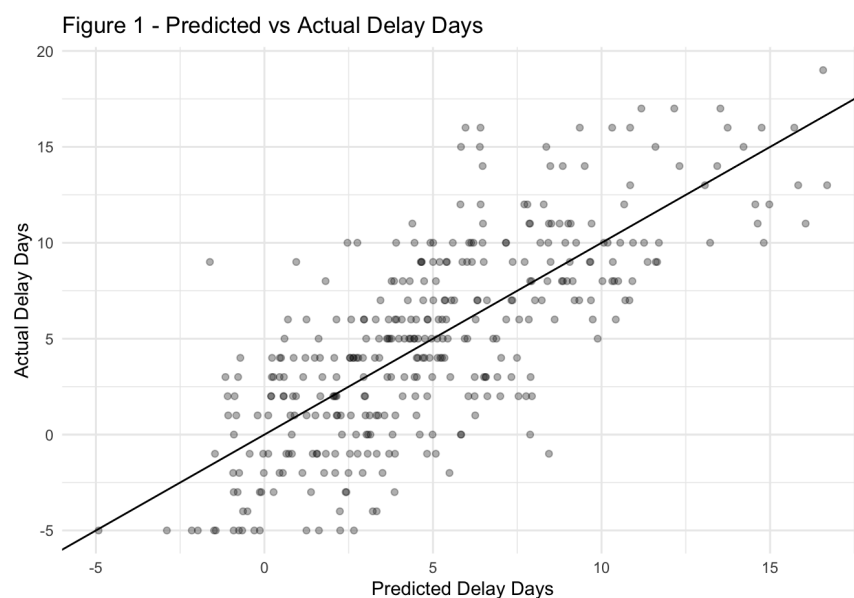


Figure 1. Predicted vs Actual Delay Days

This plot compares the model's predicted delay days against the actual delay days in the test dataset. Points that fall close to the diagonal reference line represent more accurate predictions. The plot shows that the model captures the general pattern of supplier delay risk, but there is still noticeable variation around the line.

Overall, the linear regression model is useful because it gives managers a numeric estimate of how late or early an order is likely to arrive. This can help production planners decide whether to adjust schedules, follow up with suppliers, increase safety stock, or prepare backup options.

5. Logistic Regression Results

The logistic regression model was built to predict Delay_Status, which classifies each purchase order as either Delayed or On_Time. Since the target variable is categorical, logistic regression was used. The model used a 70/30 train-test split, and was trained on 840 observations and tested on 360 observations. On_Time was used as the reference class, which means the model estimated the probability that a purchase order would be delayed.

The logistic regression model achieved an accuracy of about 85.28% on the test data. This was better than the baseline accuracy of about 79.44%, which would be achieved by predicting every test observation as the majority class, Delayed. The model improved accuracy by about 5.84 percentage points over this simple baseline.

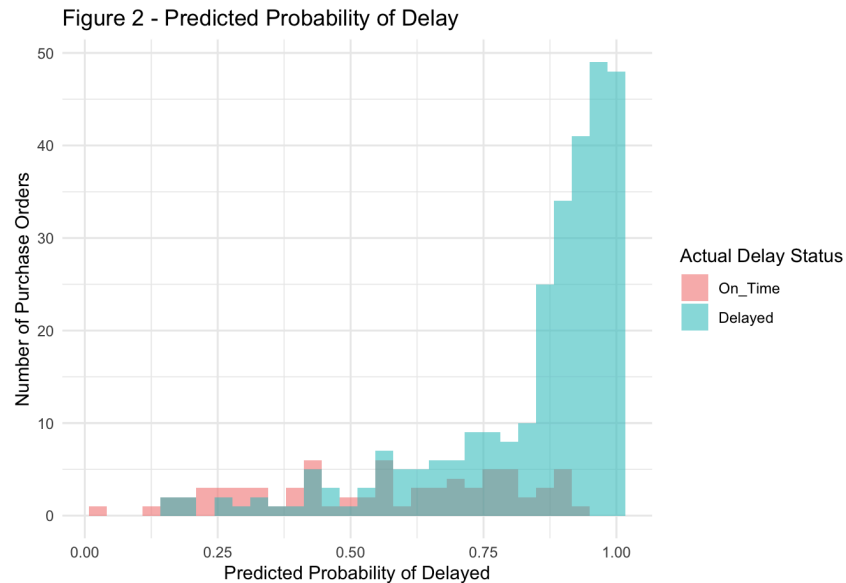


Figure 2. Predicted Probability of Delay

This plot shows the distribution of predicted delay probabilities for orders that were actually delayed and orders that were actually on time. In this analysis, many delayed orders received high predicted probabilities, which shows that the model is effective at identifying high risk purchase orders. However, some on-time orders also received high delay probabilities, which explains why the model sometimes incorrectly flags on-time orders as delayed.

The model also had an AUC of about 0.868, which suggests that it does a good job separating delayed and on-time orders based on predicted probabilities. Important variables in the logistic regression model included rush order status, supplier region, lead time, material criticality, season, and distance. Rush orders had much higher odds of being delayed, while some supplier regions had lower odds of delay. Summer and winter were also associated with higher delay risk.

6. Managerial Recommendations

The first recommendation is to use the logistic regression model as an early warning system for supplier delays. Since the model predicts the probability that a purchase order will be delayed, managers can use it to identify high risk orders before the expected delivery date. Orders with high predicted delay probabilities should receive earlier supplier follow-up, closer tracking, and backup planning.

The second recommendation is to use the linear regression model for production planning. The model's average error is about 2.8 days, so it should not be treated as an exact forecast. However, it is useful for estimating whether an order is likely to be slightly early, slightly late, or several days late. Production planners can use this estimate to decide whether to adjust schedules, delay production steps, or increase short term inventory buffers

Lastly, managers should continue monitoring supplier reliability metrics. The descriptive analysis showed that delayed orders had lower supplier ratings, higher prior delay rates, lower on-time rates, higher defect rates, and lower quality audit scores. These measures should be included in supplier scorecards and used during supplier reviews. Suppliers with weak reliability and quality performance should receive corrective action plans or reduced order volume if performance does not improve.